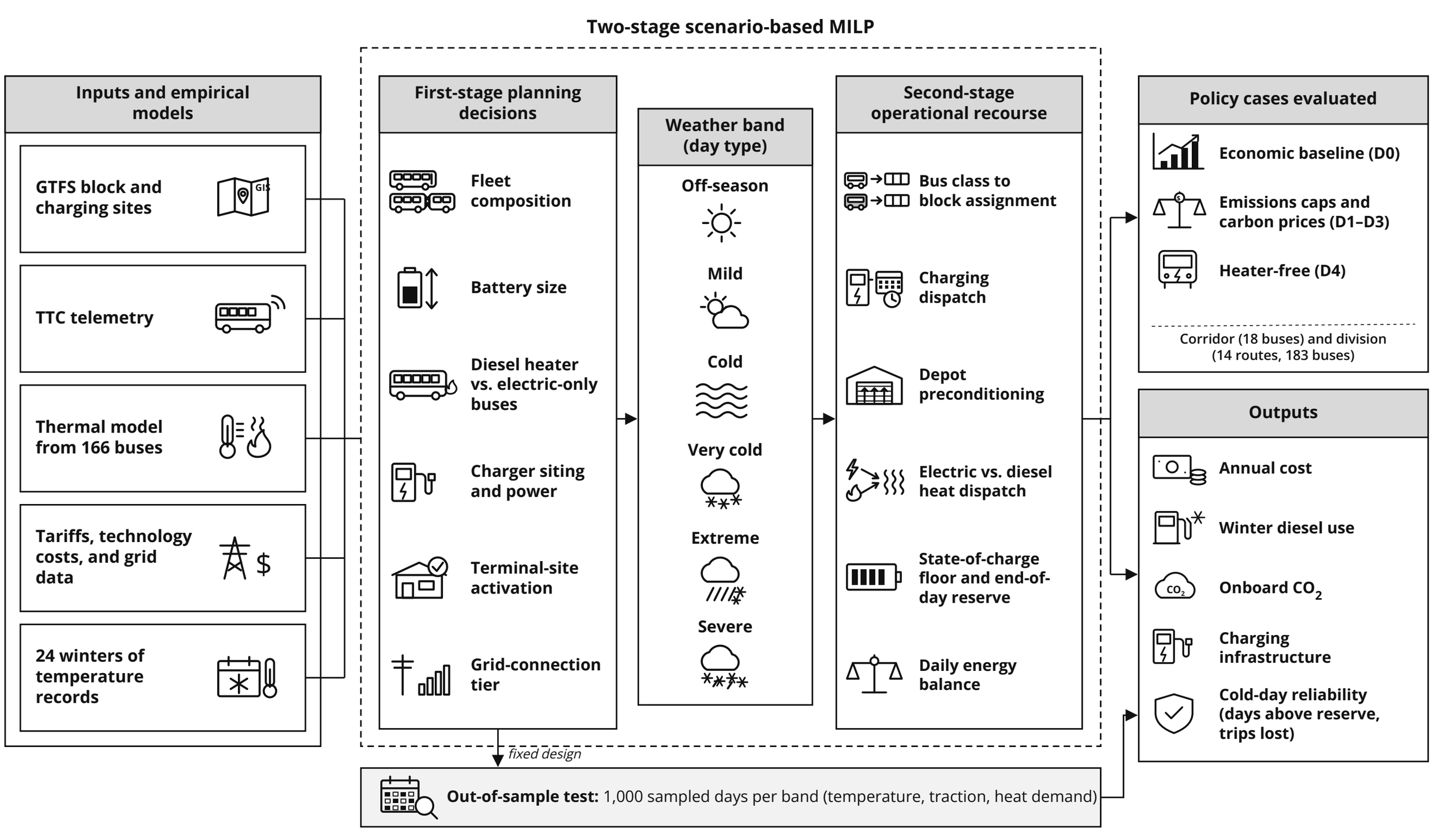

Two-stage scenario-based MILP
Inputs and empirical models
GTFS block and charging sites
GIS
TTC telemetry
Thermal model from 166 buses
Tariffs, technology costs, and grid data
$
24 winters of temperature records
First-stage planning decisions
Fleet composition
Battery size
Diesel heater vs. electric-only buses
Charger siting and power
Terminal-site activation
Grid-connection tier
Weather band (day type)
Off-season
Mild
Cold
Very cold
Extreme
Severe
Second-stage operational recourse
Bus class to block assignment
Charging dispatch
Depot preconditioning
Electric vs. diesel heat dispatch
State-of-charge floor and end-of-day reserve
Daily energy balance
fixed design
Out-of-sample test: 1,000 sampled days per band (temperature, traction, heat demand)
Policy cases evaluated
Economic baseline (D0)
Emissions caps and carbon prices (D1–D3)
Heater-free (D4)
Corridor (18 buses) and division (14 routes, 183 buses)
Outputs
Annual cost
Winter diesel use
Onboard CO_2
Charging infrastructure
Cold-day reliability (days above reserve, trips lost)